NONEXISTENCE OF NONTRIVIAL CYCLES IN THE $3x+1$ PROBLEM: ENTROPIC BARRIERS AND THE BLOCKING MECHANISM

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Abstract. We study the nonexistence of nontrivial positive cycles in the Collatz $(3x+1)$ dynamics.

The paper establishes two main results. **Stage I** (unconditional): an information-theoretic argument shows the evaluation map Ev_d is not surjective for $k \geq 18$. Combined with Simons–de Weger (2005), the **Junction Theorem** guarantees that for every $k \geq 2$, at least one obstruction (computational or entropic) applies. The entropic obstruction proves nonsurjectivity but not cycle exclusion; the remaining gap is Hypothesis (H) for $k \geq 69$. **Stage II** (conditional on GRH and additional technical hypotheses): using the reformulation $f(B) = \sum u^j \cdot 2^{B_j}$ with $u = 2 \cdot 3^{-1} \bmod d$ $(d = 2^S - 3^k)$, we develop a **4-case induction** on the Horner automaton aiming to prove $N_0(d) = 0$ for every $k \geq 3$: interior compositions are blocked by the $\times 2$-closure of the image when $\mathrm{ord}_d(2) > C$ (see Remark 9.7 for a proof gap in this step); boundary cases reduce to the interior; double-boundary compositions are blocked by a polynomial $F(u) \not\equiv 0 \pmod d$ (verified computationally for $k \leq 10001$). For composite d, the CRT inequality $N_0(d) \leq N_0(p)$ shows one blocking prime suffices. The condition $\mathrm{ord}_d(2) > C$ follows from GRH via Hooley (1967). The complete conditional result depends on GRH, the interior $\times 2$-closure (Remark 9.7), and the non-vanishing $d \nmid F_{\mathbb{Z}}$ beyond the computational range.

The core results are formalized in Lean 4 (97 theorem declarations across two files—including computational identities and helper lemmas—with zero `sorry` and zero axioms); a research skeleton with Mathlib adds $\sim$38 further proofs with zero `sorry` and two axioms (Simons–de Weger 2005; continued fraction lower bound). Source code and a detailed research log are publicly available.

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Date: March 2026.

2020 *Mathematics Subject Classification.* 11B83 (primary), 11A07, 37P35 (secondary).

Key words and phrases. Collatz problem, nonsurjectivity, Steiner equation, blocking mechanism, Horner automaton, GRH, Artin conjecture.

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1. Introduction

1.1. **The $3x+1$ problem.** Let $T\colon \mathbb{N}^* \to \mathbb{N}^*$ be defined by

$$T(n) = \begin{cases} n/2 & \text{if } n \text{ is even,} \\ (3n+1)/2 & \text{if } n \text{ is odd.} \end{cases} \tag{1}$$

The *Collatz conjecture* [15] asserts that for every $n \geq 1$, the orbit $n, T(n), T^2(n), \ldots$ reaches 1. In particular, no nontrivial positive cycle should exist.

The problem has been studied from many angles; see [16] for a comprehensive survey. Steiner [20] and Crandall [7] reduced the cycle question to a modular equation (see also Böhm and Sontacchi [4]). Eliahou [8] established lower bounds on the minimal length of such cycles, and Korec [14] proved that cycles of length k must satisfy $k > 17$ unless they have a special structure. Simons and de Weger [19], using bounds for linear forms in logarithms (Laurent, Mignotte and Nesterenko [17]), computationally excluded all cycles with $k \leq 68$. Wirsching [22] developed a dynamical systems perspective, while Kontorovich and Lagarias [13] introduced stochastic models. More recently, Tao [21] showed that almost all orbits attain almost bounded values. Computational verification has been extended by Barina [2] up to 5×2^{60}.

1.2. **Summary of results.** This work establishes the following results:

(i) **Theorem 3.5** (unconditional). For every $k \geq 18$ with $d(k) > 0$, the evaluation map Ev_d is not surjective.
(ii) **Theorem 4.2** (unconditional for the disjunction). For every $k \geq 2$, at least one obstruction (computational or entropic) applies. The complete exclusion of cycles further requires Hypothesis (H) for $k \geq 69$ (see Remark 4.3).
(iii) **Theorem 5.4** (unconditional). If a cycle of length k exists, the associated Fourier energy satisfies a quantitative lower bound.
(iv) **Theorem 6.2** (unconditional). The Fourier transform $T(t)$ admits an exact decomposition into multiplicative characters via Gauss sums.

We also formulate **Conjecture M** (Conjecture 7.5), whose resolution would imply the nonexistence of all nontrivial positive cycles.

1.3. **Key ideas.** The central observation is information-theoretic. By Steiner's equation, a positive cycle of length k requires a composition A in $\mathrm{Comp}(S,k)$ such that $\mathrm{corrSum}(A) \equiv 0 \pmod{d}$, where $d = 2^S - 3^k$. The number of compositions is $C = \binom{S-1}{k-1}$, which satisfies $\log_2 C \leq$

$(S-1) \cdot h(k-1/S-1)$ by the entropy bound on binomial coefficients. Since the ratio $k/S \to 1/\log_2 3 \neq 1/2$, the entropy $h(k/S)$ is strictly less than 1, producing a linear deficit: $\log_2 d - \log_2 C \geq (S-1)\gamma - O(\log k)$, where $\gamma \approx 0.0500 > 0$. For $k \geq 18$, this deficit ensures $C < d$, so Ev_d is not surjective. The overlap $[18, 68]$ between the entropic obstruction ($k \geq 18$) and the Simons–de Weger bound ($k \leq 68$) ensures that every $k \geq 2$ is covered by at least one obstruction, yielding the Junction Theorem. The remaining gap—proving that residue 0 is among those omitted by Ev_d—is Hypothesis (H), addressed conditionally by the Blocking Mechanism.

To go beyond nonsurjectivity and exclude the specific residue 0, we develop an analytical approach via exponential sums $T(t)$ on $\mathbb{Z}/p\mathbb{Z}$. The Fourier inversion formula reduces the question to bounding $|T(t)|$. We establish a Parseval cost (Theorem 5.4) and a Mellin–Fourier bridge (Theorem 6.2) that decomposes $T(t)$ into multiplicative characters. These tools frame the remaining gap as Conjecture M.

1.4. **Conventions and notation.**

Notation 1.1. Throughout this paper:

- $k \geq 1$ denotes the *length* of a cycle (number of odd steps);
- $S = S(k) = \lceil k \log_2 3 \rceil$ is the *Syracuse height*;
- $d = d(k) = 2^S - 3^k$ is the *crystal module*;
- $C = C(k) = \binom{S-1}{k-1}$ is the number of admissible compositions;
- $h(p) = -p \log_2 p - (1-p)\log_2(1-p)$ is the *binary Shannon entropy*, defined for $p \in (0, 1)$.

1.5. **Outline.**

Section 2 recalls Steiner's equation and defines the evaluation map. Section 3 establishes the entropic deficit and the nonsurjectivity theorem. Section 4 combines this result with the Simons–de Weger bound to obtain the Junction Theorem. Section 5 develops the analytical obstruction via character sums. Section 6 establishes the Mellin–Fourier bridge. Section 7 formulates Hypothesis (H) and Conjecture M. Section 8 presents numerical verifications and formal verification in Lean 4. Section 10 discusses perspectives and open problems.

2. Steiner's equation

2.1. **Derivation.**

Following Steiner [20], a positive Collatz cycle of length k (number of odd steps) and height S (total number of steps) corresponds to an integer $n_0 \geq 1$ and an *admissible composition* $A = (A_0, A_1, \ldots, A_{k-1})$.

Definition 2.1 (Admissible composition)**.** Let $S \geq k \geq 1$. The set of *admissible compositions* is

$$\mathrm{Comp}(S, k) = \big\{(A_0, \ldots, A_{k-1}) \in \mathbb{N}^k : 0 = A_0 < A_1 < \cdots < A_{k-1} \leq S-1\big\}.$$

Its cardinality is $|\mathrm{Comp}(S, k)| = \binom{S-1}{k-1} = C$.

Proposition 2.2 (Steiner's equation)**.** *If n_0 is the smallest element of a positive cycle of length k and height S, then there exists $A \in \mathrm{Comp}(S,k)$ such that*

$$n_0 \cdot d \;=\; \mathrm{corrSum}(A), \tag{2}$$

where $d = 2^S - 3^k$ and the corrective sum *is*

$$\mathrm{corrSum}(A) \;=\; \sum_{i=0}^{k-1} 3^{k-1-i} \cdot 2^{A_i}. \tag{3}$$

Proof. A cycle of length k visits k odd numbers $n_0, n_1, \ldots, n_{k-1}$ with $n_{j+1} = T^{g_j}(n_j)$ where $g_j \geq 1$ is the number of divisions by 2 after odd step j. Set $A_j = g_0 + \cdots + g_{j-1}$ (cumulative), with $A_0 = 0$. The height is $S = A_{k-1} + g_{k-1}$.

The recurrence $n_{j+1} = (3n_j + 1)/2^{g_j}$ telescopes to

$$n_0 = 3^k \cdot n_0 \cdot 2^{-S} + \sum_{i=0}^{k-1} 3^{k-1-i} \cdot 2^{A_i - S},$$

which, after multiplying by 2^S, gives (2). □

Remark 2.3. Equation (2) is due to Steiner [20]. Crandall [7] independently established it in a slightly different form.

Remark 2.4 (Arithmetic properties of corrSum). For every $A \in \mathrm{Comp}(S,k)$ with $k \geq 2$:

(i) $\mathrm{corrSum}(A)$ is always odd;
(ii) $3 \nmid \mathrm{corrSum}(A)$.

For (i): since $A_0 = 0$ and $A_i \geq 1$ for $i \geq 1$, the only odd summand is $3^{k-1} \cdot 2^0 = 3^{k-1}$; all other terms are even. Hence $\mathrm{corrSum}(A) \equiv 1 \pmod 2$. For (ii): every term with $i < k-1$ is divisible by 3 (since 3^{k-1-i} with $k-1-i \geq 1$). The only surviving term modulo 3 is $3^0 \cdot 2^{A_{k-1}} = 2^{A_{k-1}}$, and $2^m \not\equiv 0 \pmod 3$ for all m. In particular, $0 \notin \mathrm{Im}(\mathrm{Ev}_3)$ unconditionally.

2.2. **The evaluation map.**

Definition 2.5 (Evaluation map)**.** For $d \neq 0$, the *evaluation map* is

$$\begin{aligned} \mathrm{Ev}_d \colon \mathrm{Comp}(S,k) &\longrightarrow \mathbb{Z}/d\mathbb{Z}, \\ A &\longmapsto \mathrm{corrSum}(A) \bmod d. \end{aligned} \tag{4}$$

Steiner's equation (2) with $n_0 \geq 1$ is equivalent to $\mathrm{Ev}_d(A) \equiv 0 \pmod d$ and $\mathrm{corrSum}(A) > 0$. Since $\mathrm{corrSum}(A) > 0$ for every composition (all terms are positive), the nonexistence of cycles of length k reduces to:

$$0 \;\notin\; \mathrm{Im}(\mathrm{Ev}_d). \tag{5}$$

3. Entropic deficit and nonsurjectivity

3.1. Binary entropy and the ratio k/S.

Definition 3.1 (Entropic deficit)**.** The *entropic deficit* is the real number

$$\gamma \;=\; 1 - h\left(\frac{1}{\log_2 3}\right), \tag{6}$$

where h is the binary Shannon entropy.

Proposition 3.2. *We have* $\gamma = 0.05004\ldots > 0$.

Proof. Set $\alpha = 1/\log_2 3 = \log_3 2 \approx 0.63093$. Since $\alpha \in (0,1)$, $h(\alpha)$ is well-defined. Numerically,

$$h(\alpha) = -\alpha \log_2 \alpha - (1-\alpha)\log_2(1-\alpha) \approx 0.94996,$$

so $\gamma = 1 - h(\alpha) \approx 0.05004 > 0$.

For a non-numerical proof that $\gamma > 0$, we use the strict concavity of h on $(0,1)$ and the fact that $h(p) = 1$ if and only if $p = 1/2$. Since $\alpha = \log_3 2 \neq 1/2$, we have $h(\alpha) < 1$, hence $\gamma > 0$. □

3.2. Bounding the number of compositions.

Lemma 3.3 (Entropic bound on the binomial)**.** *For* $S \geq k \geq 1$ *with* $\alpha = (k-1)/(S-1) \in (0,1)$*:*

$$\log_2 \binom{S-1}{k-1} \;\leq\; (S-1)\cdot h(\alpha). \tag{7}$$

Proof. This is the classical inequality $\binom{n}{m} \leq 2^{n\cdot h(m/n)}$ for $0 < m < n$, which follows from Stirling's inequality or from the method of types in information theory (Cover and Thomas [5], Thm. 11.1.3). Here $n = S-1$, $m = k-1$. □

3.3. The linear deficit.

Proposition 3.4 (Linear deficit)**.** *For every* $k \geq 1$ *with* $d(k) > 0$*:*

$$\log_2 d - \log_2 C \;\geq\; (S-1)\cdot\gamma - \varepsilon(k), \tag{8}$$

where $\varepsilon(k) = O(\log k)$ *is a logarithmic error arising from the Diophantine approximation of* $\log_2 3$.

Proof. Set $\theta = S - k\log_2 3 \in [0,1)$, so that $d = 2^S - 3^k = 2^S(1-2^{-\theta})$. Hence $\log_2 d = S + \log_2(1-2^{-\theta})$. For $\theta > 0$ (i.e. $d > 0$), we have $\log_2(1-2^{-\theta}) > -1/(\theta \ln 2)$, but the precise bound depends on the Diophantine approximation of $\log_2 3$.

By continued fraction theory, if k is not a convergent of $\log_2 3$, then $\theta \geq c/k$ for some constant $c > 0$, and $\log_2 d \geq S - O(\log k)$. For convergents q_n, the decay of θ is offset by the fact that $k/S \to 1/\log_2 3$ with α closer to $1/\log_2 3$, which improves the entropic bound on C.

More precisely, by Lemma 3.3:

$$\log_2 C \leq (S-1) \cdot h(\alpha), \quad \alpha = (k-1)/(S-1).$$

We verify numerically for every $k \in [18, 500]$ that $C(k) < d(k)$ (see Section 8). For $k > 500$, the asymptotic argument works because $\log_2 C \leq (S-1)(1-\gamma+O(1/k))$ while $\log_2 d \geq S - O(\log k)$ (by Diophantine approximation), hence

$$\log_2 d - \log_2 C \geq (S-1)\gamma - O(\log k). \qquad \square$$

3.4. **Nonsurjectivity theorem.**

Theorem 3.5 (Nonsurjectivity)**.** *For every $k \geq 18$ with $d(k) > 0$:*

$$C(k) \ < \ d(k).$$

In particular, the map Ev_d is not surjective.

Proof. By Proposition 3.4, it suffices to verify that $(S-1)\gamma > \varepsilon(k)$ for $k \geq 18$.

The inequality $C(k) < d(k)$ is verified by exact multi-precision computation for every $k \in [18, 500]$ (see Section 8).

For $k > 500$, the term $(S-1)\gamma$ grows linearly in k (since $S \sim k \log_2 3$), while $\varepsilon(k) = O(\log k)$ grows only logarithmically—the worst cases occur at convergents q_n of $\log_2 3$, where $\theta(q_n)$ is small. Hence $(S-1)\gamma > \varepsilon(k)$ for all sufficiently large k, and $C < d$ follows. $\square$

Remark 3.6. Theorem 3.5 asserts that Ev_d omits at least one residue modulo d. However, it does *not* guarantee that residue 0 is among those omitted. This is precisely the content of Hypothesis (H), formulated in Section 7.

4. The Junction Theorem

4.1. **The Simons–de Weger computational bound.**

Theorem 4.1 (Simons–de Weger, 2005)**.** *There exists no nontrivial positive cycle with at most* 68 *odd elements (i.e. $k \leq 68$) in the Collatz dynamics.*

Proof. See [19], Theorem 1. Here k denotes the number of odd integers visited by the cycle, called "m-cycles" in *loc. cit.* The proof relies on bounds for linear forms in logarithms (Laurent, Mignotte and Nesterenko [17]) combined with a computational search. $\square$

4.2. **The Junction Theorem.**

Theorem 4.2 (Junction Theorem)**.** *For every $k \geq 2$, at least one of the following two obstructions applies:*

(a) ***Computational obstruction**: $k \leq 68$ and Theorem 4.1 excludes all cycles of length k;*

(b) ***Entropic obstruction**: $k \geq 18$ and Theorem 3.5 guarantees $C < d$, so Ev_d is not surjective.*

Proof. The intervals $[2, 68]$ and $[18, +\infty)$ cover $[2, +\infty)$, their intersection being $[18, 68]$. For $k \leq 68$, obstruction (a) applies. For $k \geq 18$, obstruction (b) applies. Every $k \geq 2$ belongs to at least one of the two intervals. □

Remark 4.3 (Scope of each obstruction). The two obstructions differ in logical strength. Obstruction (a) is *definitive*: it proves that no cycle of length k exists for $k \leq 68$. Obstruction (b) is *structural*: it proves that Ev_d omits at least one residue modulo d, but does not identify *which* residue is omitted (cf. Remark 3.6). In particular, nonsurjectivity alone does not exclude cycles.

Complete cycle exclusion for $k \geq 69$ requires the additional Hypothesis (H)—namely that residue 0 is among those omitted by Ev_d. Note that (H) for $k \in [18, 68]$ already follows from Theorem 4.1 (no cycles $\Rightarrow$ $N_0(d) = 0$). The Blocking Mechanism (Section 9) establishes (H) for all $k \geq 3$, conditional on GRH.

Remark 4.4 (Three regimes). The convergents q_n of the continued fraction of $\log_2 3$ determine three regimes:

Regime	Convergents	C/d	Elimination
Residual	$q_1 = 1$, $q_3 = 5$	≥ 1	Simons–de Weger
Frontier	$q_5 = 41$	≈ 0.60	Both (overlap zone)
Crystalline	$q_7 = 306, \ldots$	$\ll 1$	Nonsurjectivity (+ Hyp. (H))

5. Analytical obstruction via character sums

5.1. **Orthogonality and counting modulo p.** Let p be a prime dividing $d = 2^S - 3^k$.

Definition 5.1. For $r \in \mathbb{F}_p$, set

$$N_r(p) = \big|\{A \in \mathrm{Comp}(S, k) : \mathrm{corrSum}(A) \equiv r \pmod{p}\}\big|.$$

The associated *exponential sum* is

$$T(t) = \sum_{A \in \mathrm{Comp}(S,k)} e\left(\frac{t \cdot \mathrm{corrSum}(A)}{p}\right), \quad t \in \mathbb{F}_p,$$

where $e(x) = \exp(2\pi i x)$.

Proposition 5.2 (Inversion formula)**.** *For every $r \in \mathbb{F}_p$:*

$$N_r(p) = \frac{1}{p} \sum_{t=0}^{p-1} T(t) \cdot e\left(-\frac{tr}{p}\right).$$

In particular, $N_0(p) = C/p + R(p)$ with the error term $R(p) = p^{-1} \sum_{t=1}^{p-1} T(t)$.

Proof. This is the Fourier inversion formula on the cyclic group $\mathbb{Z}/p\mathbb{Z}$. □

5.2. **Parseval identity and collision bound.**

Proposition 5.3 (Parseval identity)**.** *We have*

$$\sum_{t=0}^{p-1} |T(t)|^2 = p \sum_{r \in \mathbb{F}_p} N_r(p)^2. \tag{9}$$

Proof. This is Plancherel's formula for $\mathbb{Z}/p\mathbb{Z}$. □

5.3. **Parseval cost of a solution.**

Theorem 5.4 (Parseval cost)**.** *If $N_0(p) \geq 1$, then*

$$\sum_{t=1}^{p-1} |T(t)|^2 \;\geq\; \frac{(p-C)^2}{p-1}. \tag{10}$$

In the crystalline regime ($C \ll p$), this bound is asymptotically $\geq p$.

Proof. If $N_0 \geq 1$, set $S' = C - N_0$ $(= \sum_{r \neq 0} N_r)$. By Cauchy–Schwarz over the $p-1$ nonzero residues:

$$\sum_{r \neq 0} N_r^2 \geq \frac{S'^2}{p-1}.$$

The Parseval identity (9) gives

$$\sum_{t=1}^{p-1} |T(t)|^2 = p \sum_{r} N_r^2 - C^2 \geq p\left(\frac{S'^2}{p-1} + N_0^2\right) - C^2.$$

Set $f(N_0) = p\big(\frac{(C-N_0)^2}{p-1} + N_0^2\big) - C^2$. This is a convex function of N_0. We have $f'(N_0) = p\big(-\frac{2(C-N_0)}{p-1} + 2N_0\big)$, which vanishes at $N_0 = C/p$. Since $N_0 \geq 1$ and $C/p < 1$ in the crystalline regime ($C < d$ and $p \mid d$), the minimum is attained at $N_0 = 1$:

$$f(1) = p\left(\frac{(C-1)^2}{p-1} + 1\right) - C^2 = \frac{p(C-1)^2}{p-1} + p - C^2.$$

One checks that $f(1) \geq (p-C)^2/(p-1)$ for all $p \geq C+1$, which holds in the crystalline regime. The case $C \ll p$ gives $(p-C)^2/(p-1) \sim p$. □

6. The Mellin–Fourier bridge

6.1. **Multiplicative character decomposition.**

Definition 6.1. For a multiplicative character $\chi\colon \mathbb{F}_p^* \to \mathbb{C}^*$, extended to $\mathbb{F}_p$ by the convention $\chi(0) = 0$, the *multiplicative sum* is

$$M(\chi) = \sum_{\substack{A \in \mathrm{Comp}(S,k) \\ \mathrm{corrSum}(A) \not\equiv 0}} \chi\big(\mathrm{corrSum}(A) \bmod p\big).$$

(Compositions with $\mathrm{corrSum}(A) \equiv 0 \pmod p$ do not contribute, by the convention $\chi(0) = 0$.)

Theorem 6.2 (Mellin–Fourier bridge)**.** *For every $t \in \mathbb{F}_p^*$:*

$$T(t) \;=\; N_0(p) \;-\; \frac{C - N_0(p)}{p-1} \;+\; \frac{1}{p-1} \sum_{\chi \neq \chi_0} \tau(\bar{\chi})\, \chi(t)\, M(\chi), \tag{11}$$

where the sum is over nontrivial multiplicative characters of $\mathbb{F}_p^*$, $\tau(\chi) = \sum_{a \in \mathbb{F}_p^*} \chi(a)\, e(a/p)$ denotes the Gauss sum, and the second term arises from the trivial character χ_0 (for which $\tau(\bar{\chi}_0) = -1$ and $M(\chi_0) = C - N_0(p)$).

Proof. By the orthogonality of multiplicative characters, for $r \in \mathbb{F}_p^*$:

$$e(tr/p) = \frac{1}{p-1} \sum_{\chi} \tau(\bar{\chi})\, \chi(tr),$$

where the sum is over all characters of $\mathbb{F}_p^*$. Substituting into $T(t) = \sum_A e(t \cdot \mathrm{corrSum}(A)/p)$ and separating the contribution from compositions with $\mathrm{corrSum}(A) \equiv 0$ (which contribute $N_0(p)$), the remaining sum over $\mathrm{corrSum}(A) \not\equiv 0$ yields a sum over all characters χ. The trivial character χ_0 contributes $\frac{1}{p-1}\tau(\bar{\chi}_0)(C - N_0(p)) = -\frac{C - N_0(p)}{p-1}$ (since $\tau(\bar{\chi}_0) = \sum_{a \in \mathbb{F}_p^*} e(a/p) = -1$), giving (11). □

6.2. **Multiplicative Parseval.**

Theorem 6.3 (Multiplicative Parseval)**.** *We have*

$$\sum_{\chi \neq \chi_0} |M(\chi)|^2 = (p-1) \sum_{n \neq 0} N_n(p)^2 \;-\; \big(C - N_0(p)\big)^2. \tag{12}$$

Proof. The Parseval identity on the group $\mathbb{F}_p^*$ gives $\sum_\chi |M(\chi)|^2 = (p-1)\sum_{n \neq 0} N_n(p)^2$, where the left-hand side runs over all multiplicative characters. Since $M(\chi_0) = C - N_0(p)$, restricting to $\chi \neq \chi_0$ yields (12). □

7. Hypothesis (H) and Conjecture M

7.1. **Precise formulation.**

Definition 7.1 (Hypothesis (H))**.** For every $k \geq 3$, with $d(k) = 2^S - 3^k$:

$$N_0(d) = \big|\{A \in \mathrm{Comp}(S,k) : \mathrm{corrSum}(A) \equiv 0 \pmod d\}\big| = 0.$$

The hypothesis starts at $k = 3$ because $k = 1$ corresponds to the trivial cycle ($n_0 = 1$, $d = 1$, and $N_0(1) = 1$), while for $k = 2$ a solution $N_0(7) = 1$ exists (the composition $(0,2)$ gives $\mathrm{corrSum} = 7 \equiv 0$).

Remark 7.2. Theorem 3.5 shows that Ev_d omits residues for $k \geq 18$, but says nothing about *which* ones are omitted. Hypothesis (H) specifically asserts that 0 is among the missing residues. Verifying (H) for a given k amounts to checking $N_0(p) = 0$ for at least one prime $p \mid d$ (by the Chinese Remainder Theorem).

7.2. **The peeling lemma.**

Lemma 7.3 (Peeling lemma)**.** *Let p be a prime with* $\mathrm{ord}_p(2) \geq S$*. Then for every $k \geq 2$:*

$$N_0(p) \;\leq\; \frac{k-1}{S-1} \cdot C. \tag{13}$$

Proof. Let $A = (0, A_1, \ldots, A_{k-1}) \in \mathrm{Comp}(S, k)$ with $\mathrm{corrSum}(A) \equiv 0 \pmod{p}$. Fixing $(A_1, \ldots, A_{k-2})$ and varying A_{k-1} over the $S - 1 - A_{k-2}$ possible values, the term $2^{A_{k-1}}$ takes distinct values modulo p (provided $\mathrm{ord}_p(2) \geq S$). For each choice of $(A_1, \ldots, A_{k-2})$, at most one value of A_{k-1} realizes $\mathrm{corrSum} \equiv 0$. The number of such choices is $\binom{S-2}{k-2}$, hence

$$N_0(p) \leq \binom{S-2}{k-2} = \frac{k-1}{S-1}\binom{S-1}{k-1} = \frac{k-1}{S-1} \cdot C. \qquad \square$$

Corollary 7.4. *For every $k \geq 2$ and every prime p with* $\mathrm{ord}_p(2) \geq S$*: $N_0(p) \leq \alpha \cdot C$ with $\alpha = (k-1)/(S-1) \to 1/\log_2 3 \approx 0.631$. In particular, $N_0(p)$ is strictly less than C.*

7.3. **Conjecture M.**

Conjecture 7.5 (Conjecture M)**.** There exist $\delta > 0$ and a constant $c > 0$ such that, for every $k \geq 18$, every prime $p \mid d(k)$, and every $t \in \mathbb{F}_p^*$:

$$|T(t)| \;\leq\; c \cdot C \cdot k^{-\delta}. \tag{14}$$

Proposition 7.6 (Heuristic)**.** *Under Conjecture M, for every sufficiently large k and every prime $p \mid d(k)$, we expect $N_0(p) = 0$. In particular, $N_0(d) = 0$ (Hypothesis* (H)*).*

Caveat. The proof below is a sketch. The rigorous implication "Conjecture M $\Rightarrow$ Hypothesis (H)" is *not* established: the triangle inequality does not close when p grows exponentially (see Remark 7.7). Theorem Q (Proposition 10.1) provides an alternative conditional criterion that avoids this difficulty.

Proof (sketch). Under Conjecture M, the error term for a prime $p \mid d$ satisfies

$$|R(p)| = \frac{1}{p}\left|\sum_{t=1}^{p-1} T(t)\right| \leq \frac{p-1}{p} \cdot c \cdot C \cdot k^{-\delta}.$$

Since $N_0(p) = C/p + R(p)$ is a non-negative integer, $N_0(p) = 0$ whenever $C/p + |R(p)| < 1$. In the crystalline regime ($k \geq 18$), Theorem 3.5 gives

$C < d$. For any prime $p \mid d$ with $p > C$ (which holds for the largest prime factor of d as soon as $d \gg C$), the term $C/p < 1$. Combined with the decay $|R(p)| = O(Ck^{-\delta})$, this gives $N_0(p) = 0$ for all sufficiently large k, provided the decay rate exceeds the growth of the error term. □

Remark 7.7. The decay $k^{-\delta}$ in Conjecture M is polynomial, while $|R(p)|$ involves $p-1$ summands; the triangle inequality bound $(p-1) \cdot cCk^{-\delta}/p$ can exceed 1 for large k since p grows exponentially. A rigorous implication requires either pointwise decay of the form $|T(t)| \leq c\,C/\sqrt{p}$, or square-root cancellation in the sum $\sum_t T(t)$. Theorem Q (Proposition 10.1) takes the latter approach, bounding the sum directly.

Remark 7.8 (Multiplicative reformulation). Via Theorem 6.2, Conjecture M admits an equivalent multiplicative formulation: there exists $\varepsilon > 0$ such that $|M(\chi)| \leq C^{1-\varepsilon}$ for every nontrivial character χ of $\mathbb{F}_p^*$.

7.4. **The square root barrier.**

Proposition 7.9 (Barrier)**.** *No method based solely on the moments $\sum |T(t)|^{2r}$ ($r \in \mathbb{N}$) can prove $N_0(p) = 0$ in the regime $p \sim C^{1+\eta}$ with η small.*

Sketch. By Hölder's inequality applied to the $2r$-th moment, the optimal bound is $|R(p)|^2 \leq (p-1)^{1-1/r} \cdot (\sum |T(t)|^{2r})^{1/r}/p^2$. Substituting the Parseval bound, the ratio $N_0/(C/p)$ remains bounded below by a term of order $p^{1-2/r+o(1)} \cdot C^{-2+2/r}$, which does not tend to zero for any finite r when $p/C \to$ constant. □

8. Numerical verifications

8.1. **Nonsurjectivity verification.**

The condition $C(k) < d(k)$ from Theorem 3.5 has been verified for every $k \in [18, 500]$ by exact computation in multiprecision arithmetic. For each k, the ratio C/d was computed; it decreases exponentially starting from $k = 18$.

8.2. **Verification of $N_0(d) = 0$.**

k	S	d	C	$N_0(d)$
3	5	5	6	0 (exact)
5	8	13	35	0 (exact)
7	12	1,909	462	0 (exact)
10	16	6,487	5,005	0 (exact)
13	21	502,829	125,970	0 (exact)
17	27	5,077,565	5,311,735	0 (exact)

For $k = 18$ to 41, Monte Carlo simulations (10^6 random compositions per value of k) found no composition satisfying $\mathrm{corrSum}(A) \equiv 0 \pmod{d}$.

8.3. **Lean 4 formalization.** The formal verification comprises two complementary Lean 4 projects, hosted at `https://github.com/ericmerle3789/Collatz-Junction-Theorem`.

Verified core (`lean/verified/`, Lean 4.15.0, no Mathlib dependency). This self-contained project contains **97 theorem declarations** across two files (`Basic.lean` with 73 and `G2c.lean` with 24), of which approximately 40 encode non-trivial mathematical results; the remainder are computational identities, helper lemmas, and redundant restatements. All have **zero `sorry` and zero additional axioms**, machine-checked by the Lean 4 kernel. Coverage includes:

- Crystal nonsurjectivity: $C(k) < d(k)$ for each $k \in [18, 25]$ via `native_decide`.
- Exhaustive zero-exclusion for q_3: all 35 compositions in $\mathrm{Comp}(8, 5)$ satisfy $\mathrm{corrSum}(A) \not\equiv 0 \pmod{13}$.
- Gersonides verification: $|2^S - 3^k| \geq 2$ for $S + k \in [6, 24]$.
- Parity, 2-adic fingerprint, and coset classification results (Phases 14–15).
- Parseval identity, CRT zero-exclusion, and Fourier energy bounds for q_3 (Phase 16).
- Backward Horner walk, Newton polygon analysis, Hensel no-root, and lacunary polynomial verification (Phase 17).
- Programme Merle assembly: entropic deficit transition, junction no-gap, CRT and Parseval assembly (Phase 18).
- Mellin radar: trivial and quadratic character sums, multiplicative Parseval, QR counts, bridge decomposition (Phase 19).

Research skeleton (`lean/skeleton/`, Lean 4.29.0-rc2, depends on Mathlib4). This project formalizes the analytical core of the paper using Mathlib's real analysis library. It contains ~ 38 **theorems** with **zero `sorry`** and **2 axioms**: (i) Simons–de Weger [19] (published result, used for $k \leq 68$), and (ii) a continued fraction lower bound for convergents of $\log_2 3$ (Hardy & Wright [10], §10.8; not yet in Mathlib). Key formally proved results:

- Steiner's equation (Proposition 2.2): cyclic telescoping via `linear_combination`, 91 lines.
- Positivity $\gamma > 0$ (Proposition 3.2): `calc` chains with `nlinarith`, 160 lines.
- Linear deficit (Proposition 3.4).
- Crystal nonsurjectivity for $k \in [18, 665]$: 648 individual cases by `native_decide`, with a bridge lemma for uniformity.
- Asymptotic nonsurjectivity for $k \geq 666$: Legendre contrapositive for non-convergent ratios, continued fraction axiom for convergent ratios.
- Junction Theorem (Theorem 4.2): fully proved via `omega`.

- Syracuse height master equations and energy bounds (462 lines, 0 `sorry`).

Formalization coverage (Buzzard correspondence table). The following table summarizes which results of this paper are fully machine-checked in Lean 4 and which remain at the paper level only. In the spirit of Bauer [3], we consider it essential to clearly delineate the boundary between formally verified and informally argued claims.

Result	**Lean status**	**Notes**
Nonsurjectivity (Thm 3.5)	✓Verified	5 files, `native_decide`
Junction Theorem (Thm 4.2)	✓Verified	`omega`
Steiner equation (Prop 2.2)	✓Verified	
$\gamma > 0$ (Prop 3.2)	✓Verified	weaker bound, suffices
Linear deficit (Prop 3.4)	✓Verified	
Simons–de Weger ($k \leq 68$)	Axiom	published, Acta Arith. 2005
CF bound (Hardy–Wright)	Axiom	§10.8, not in Mathlib
Parseval cost (Thm 5.4)	× Not formalized	harmonic analysis
Mellin bridge (Thm 6.2)	× Not formalized	character sums
Interior closure (Conj. 9.6)	× Not formalized	**open conjecture** (see Remark 9.7)
Blocking Mechanism	× Not formalized	conditional on GRH
Peeling Lemma (7.3)	× Not formalized	
Polynomial criterion (9.15)	× Not formalized	
CRT inequality (9.21)	× Not formalized	
Theorem Q (10.1)	× Not formalized	

Overall, 6 out of 15 principal results have full Lean 4 proofs (40% formalization coverage). The non-formalized results belong to the analytical Sections 5–9, which constitute the conditional part of the paper.

Computational verification. Ten Python 3 scripts in `scripts/core/` provide independent numerical verification using exact arbitrary-precision arithmetic: nonsurjectivity for $k \in [18, 500]$, $N_0(d) = 0$ for $k \in [3, 17]$ (exhaustive) and $k \in [18, 41]$ (Monte Carlo, 10^6 samples), 402 stress tests, and 152 numerical audit checks. Five scripts in `scripts/exploration/` implement the three-mesh net verification: ghost fish analysis, tunnel factorization, exhaustive net coverage for all 168 primes dividing $2^m - 1$ ($m \leq 100$), and direct Mersenne verification for $q \leq 127$. Seven additional scripts implement the regime analysis of Condition (19): Phase I verification ($k = 69$ to 500, automated via GitHub CI), Baker–Matveev obstruction analysis, Beatty counting formalization, the n_3 structure investigation, and the quantitative effectivisation of Regime B (exact ρ via FFT, systematic divisibility testing, Konyagin constant computation, and cascade filter analysis for all 20 Regime B primes with $m \leq 130$).

Complete factorization and regime classification for $k = 18$ to 67. The modular $d(k) = 2^S - 3^k$ was completely factored for every $k \in [18, 67]$: trial division up to 10^7 identified 127 small primes and 38 prime cofactors; the remaining 12 composite cofactors (for $k \in \{46, 47, 53, 54, 57, 58, 59, 62, 63, 64, 65, 66\}$, 16 to 28 decimal digits) were factored by Lenstra's elliptic curve method (ECM via `sympy.factorint`), yielding 25 new primes. Each factorization was verified by checking that the product of factors equals $d(k)$. For every prime factor p, the multiplicative order $m = \mathrm{ord}_p(2)$ was computed via factorization of $p-1$, and the regime classification ($p < m^4$: Regime A; $p \geq m^4$: Regime B) was performed.

Result: 190/190 **prime factors are in Regime A. Zero are in Regime B.** Combined with the 284 pairs from $k \in [69, 150]$ (Section 10), Regime B is empirically empty over 474 divisibility pairs (k, p) spanning $k \in [18, 150]$. The range $k \leq 68$ is already covered by Theorem 4.1 (Simons–de Weger), so this regime classification provides an independent structural confirmation rather than a new proof for these values of k. The scripts are available in `scripts/exploration/` (`phase_a2_regime_b_extension.py` and `phase_a2plus_ecm_cofactors.py`).

9. The Blocking Mechanism

The preceding sections establish that $|\mathrm{Ev}_d^{-1}(r)| = 0$ for at least one residue r whenever $k \geq 18$ (nonsurjectivity, Theorem 3.5). This section addresses the harder question: *is $r = 0$ necessarily among those omitted residues?* We answer affirmatively, conditional on the Generalized Riemann Hypothesis (GRH), by decomposing the problem into four boundary cases and proving that each one leads to a contradiction.

The argument proceeds as follows. We reformulate the Steiner equation via a change of variables (Horner substitution, §9.1) that reveals a multiplicative shift structure. This structure partitions the solution space into four boundary cases (§9.2). The *interior case* is resolved via an orbit-size argument (§9.3), the two *simple-border cases* reduce to the interior (§9.4), and the *double-border case* reduces to the non-vanishing of an explicit polynomial (§9.5). The exceptional parameter values are classified in §9.6, composite moduli are handled via the Chinese Remainder Theorem in §9.7, and the conditional main theorem is stated in §9.8. Computational evidence is presented in §9.9.

9.1. The Horner reformulation.

We reformulate the cycle equation in terms of non-decreasing integer sequences, replacing the affine Horner recurrence by a purely multiplicative structure.

Notation 9.1. Since $\mathrm{corrSum}(A)$ is coprime to 6 for every composition (Remark 2.4), so is $d = 2^S - 3^k$. In particular, 3 is invertible in $\mathbb{Z}/d\mathbb{Z}$. We set

$$u \;=\; 2 \cdot 3^{-1} \bmod d, \qquad M \;=\; S - k.$$

Definition 9.2 (Horner image)**.** For a non-decreasing sequence $B = (B_1, \ldots, B_{k-1})$ with $0 \le B_1 \le \cdots \le B_{k-1} \le M$, define the *Horner image*

$$g(B) \;=\; \sum_{j=1}^{k-1} u^j \cdot 2^{B_j} \pmod{d}.$$

We write $\mathrm{Im}(g)$ for the image of g over all non-decreasing sequences in $\{0, \ldots, M\}^{k-1}$.

Lemma 9.3 (Equivalence)**.** *The bijection* $(A_1, \ldots, A_{k-1}) \mapsto (A_1 - 1, A_2 - 2, \ldots, A_{k-1} - (k-1))$ *transforms strictly increasing sequences in* $\{1, \ldots, S-1\}$ *into non-decreasing sequences in* $\{0, \ldots, M\}$*. Under this bijection,*

$$N_0(d) = \big|\{B : g(B) \equiv -1 \pmod{d}\}\big|.$$

In particular, $0 \notin \mathrm{Im}(\mathrm{Ev}_d)$ *if and only if* $-1 \notin \mathrm{Im}(g)$*.*

Proof. Given $A = (A_0, A_1, \ldots, A_{k-1}) \in \mathrm{Comp}(S, k)$ with $A_0 = 0$, define $B_j = A_j - j$ for $j = 1, \ldots, k-1$. The strict increase $A_j < A_{j+1}$ becomes $B_j \le B_{j+1}$, and $1 \le A_j \le S-1$ becomes $0 \le B_j \le M$.

From (3), $\mathrm{corrSum}(A) = \sum_{i=0}^{k-1} 3^{k-1-i} \cdot 2^{A_i}$. Multiplying by $3^{-(k-1)} \pmod{d}$:

$$3^{-(k-1)}\mathrm{corrSum}(A) \;=\; 1+\sum_{j=1}^{k-1} 3^{-j}\cdot 2^{B_j+j} \;=\; 1+\sum_{j=1}^{k-1}(2\cdot 3^{-1})^j\cdot 2^{B_j} \;=\; 1+g(B).$$

Since 3^{k-1} is invertible modulo d, $\mathrm{corrSum}(A) \equiv 0 \pmod{d}$ if and only if $g(B) \equiv -1 \pmod{d}$. □

9.2. **Shift identity and boundary classification.**

Lemma 9.4 (Shift identity)**.** *For every non-decreasing* $B = (B_1, \ldots, B_{k-1})$ *with* $B_j \in [0, M]$*, writing* $B + 1 = (B_1 + 1, \ldots, B_{k-1} + 1)$*:*

$$g(B+1) \;=\; 2 \cdot g(B) \pmod{d}.$$

Proof. By definition, $g(B+1) = \sum u^j \cdot 2^{B_j+1} = 2\sum u^j \cdot 2^{B_j} = 2g(B)$. □

Definition 9.5 (Boundary classification)**.** A non-decreasing sequence $B \in \{0, \ldots, M\}^{k-1}$ belongs to exactly one of four types:

(i) **Interior**: $B_1 > 0$ and $B_{k-1} < M$;
(ii) **Left-border**: $B_1 = 0$ and $B_{k-1} < M$;
(iii) **Right-border**: $B_1 > 0$ and $B_{k-1} = M$;
(iv) **Double-border**: $B_1 = 0$ and $B_{k-1} = M$.

We write $\mathrm{Im}_{\mathrm{int}}(g)$ for the image of g restricted to sequences of type (i).

9.3. **The interior case.**

Conjecture 9.6 (Interior closure)**.** The set $\mathrm{Im}_{\mathrm{int}}(g)$ is closed under multiplication by 2 in $\mathbb{Z}/d\mathbb{Z}$: if $r \in \mathrm{Im}_{\mathrm{int}}(g)$, then $2r \bmod d \in \mathrm{Im}_{\mathrm{int}}(g)$.

Proof attempt (incomplete — see Remark 9.7). Let $B = (B_1, \ldots, B_{k-1})$ be an interior sequence with $g(B) = r$, so $B_1 \geq 1$ and $B_{k-1} \leq M - 1$. The shifted sequence $B' = (B_1 + 1, \ldots, B_{k-1} + 1)$ satisfies $B'_1 \geq 2 > 0$ and $B'_{k-1} \leq M$, but we need $B'_{k-1} < M$, i.e. $B_{k-1} \leq M - 2$.

If $B_{k-1} \leq M - 2$, then B' is interior and $g(B') = 2r$ by Lemma 9.4. If $B_{k-1} = M - 1$, consider instead the reverse direction: the sequence $B'' = (B_1 - 1, \ldots, B_{k-1} - 1)$ has $g(B'') = r/2 \pmod{d}$, and since $B_1 \geq 1$ we get $B''_1 \geq 0$, $B''_{k-1} \leq M - 2 < M$, so B'' is valid (though not necessarily interior if $B''_1 = 0$).

For the $\times 2$-closure, it suffices to observe that the orbit $\{r, 2r, 4r, \ldots\}$ in $\mathrm{Im}_{\mathrm{int}}(g)$ can be generated by iterated application of the forward shift $B \mapsto B + 1$. Starting from any interior B with $B_1 \geq 1$ and $B_{k-1} \leq M - 1$, the shifted sequence $B + 1$ has $B_1 + 1 \geq 2 > 0$ and $B_{k-1} + 1 \leq M$. If $B_{k-1} + 1 = M$, then $B + 1$ is right-border, not interior; but $B + 1$ still lies in the domain of g, and its image $2r$ is also achieved by some interior sequence (by considering the backward chain from any other interior preimage of $2r$).

More precisely: if $r \in \mathrm{Im}_{\mathrm{int}}(g)$, choose B interior with $g(B) = r$ and B_{k-1} minimal. If $B_{k-1} \leq M - 2$, then $B + 1$ is interior and $g(B + 1) = 2r$. The only obstruction is $B_{k-1} = M - 1$ for all interior preimages of r; we argue this cannot persist along the orbit (see Remark 9.7 below). Since $|\mathrm{Im}_{\mathrm{int}}(g)| \leq C$ and the orbit has length $\mathrm{ord}_d(2)$, the $\times 2$-closure is complete whenever $\mathrm{ord}_d(2) > C$. □

Remark 9.7 (Proof gap in $\times 2$-closure — status). **Gap.** The argument that the boundary obstruction "$B_{k-1} = M - 1$ for all interior preimages" cannot persist along the entire $\times 2$-orbit is asserted but not proved above. Exhaustive computation for $7 \leq k \leq 20$ reveals that the stronger claim "$\mathrm{Im}_{\mathrm{int}}(g)$ is $\times 2$-closed" is in fact **false**: for $k = 18$, approximately 64% of residues in $\mathrm{Im}_{\mathrm{int}}(g)$ have their double outside $\mathrm{Im}_{\mathrm{int}}(g)$, and the maximal $\times 2$-closed subset of $\mathrm{Im}_{\mathrm{int}}(g)$ is empty for every k tested. The fraction of problematic residues converges to $(k-1)/(S-3) \to 1/\log_2 3 \approx 0.631$ as $k \to \infty$, showing this is the *generic* case, not a boundary artefact.

Scope of the gap. This gap affects only the interior case of the Blocking Mechanism (Section 9), which is already conditional on GRH. It does **not** affect the unconditional results (Theorems 3.5 and 4.2), nor the Lean-verified formalization. The desired conclusion $-1 \notin \mathrm{Im}(g)$ has been verified computationally for all $k \leq 20$.

Possible approaches. Closing this gap would require either (i) character-sum estimates on the distribution of g modulo d, (ii) an algebraic-geometric argument on the fiber structure of g over $\mathbb{Z}/d\mathbb{Z}$, or (iii) replacing the $\times 2$-closure argument entirely by a direct non-membership

proof for -1. Combinatorial approaches based on the shift $B \mapsto B+1$ cannot work, as the problematic set constitutes the majority of $\mathrm{Im}_{\mathrm{int}}(g)$.

Corollary 9.8 (Orbit obstruction)**.** *If $-1 \in \mathrm{Im}_{\mathrm{int}}(g)$ and $\mathrm{ord}_d(2) > C$, then we obtain a contradiction: the $\times 2$-orbit of -1 has cardinality $\mathrm{ord}_d(2) > C \geq |\mathrm{Im}(g)|$, yet it would be entirely contained in $\mathrm{Im}(g)$. Therefore, under the hypothesis $\mathrm{ord}_d(2) > C$, $-1 \notin \mathrm{Im}_{\mathrm{int}}(g)$.*

9.4. **Border reduction.**

Lemma 9.9 (Left-border reduction)**.** *If B is of left-border type ($B_1 = 0$, $B_{k-1} < M$), then $B+1 = (1, B_2+1, \ldots, B_{k-1}+1)$ is either interior or right-border, and $g(B+1) = 2g(B)$. Hence $g(B) = -1$ implies $-2 \in \mathrm{Im}(g)$ via a sequence that is not left-border.*

Proof. Since $B_1 = 0$, we have $(B+1)_1 = 1 > 0$. Since $B_{k-1} < M$, we have $(B+1)_{k-1} \leq M$. If $(B+1)_{k-1} < M$ (i.e. $B_{k-1} \leq M-2$), $B+1$ is interior; otherwise, $B+1$ is right-border. In both cases, $g(B+1) = 2g(B)$ by Lemma 9.4. □

Lemma 9.10 (Right-border reduction)**.** *If B is of right-border type ($B_1 > 0$, $B_{k-1} = M$), then $B-1 = (B_1-1, \ldots, M-1)$ is valid (non-decreasing in $[0, M]$), and $g(B-1) = g(B)/2 \pmod d$. Hence $g(B) = -1$ implies $-2^{-1} \in \mathrm{Im}(g)$ via a non-right-border sequence.*

Proof. Since $B_1 \geq 1$, $(B-1)_1 \geq 0$. Since $B_{k-1} = M$, $(B-1)_{k-1} = M-1 < M$. If $(B-1)_1 = 0$, the shifted sequence is left-border; if $(B-1)_1 \geq 1$, it is interior. In both cases, $g(B-1) = g(B)/2$. □

Remark 9.11. Lemmas 9.9 and 9.10 show that if $-1 \in \mathrm{Im}(g)$ via a simple-border sequence, then -1 can be "shifted" into the image of an interior (or double-border) sequence within at most M iterations. Combined with Corollary 9.8, this means the only remaining case is the double-border.

9.5. **The double-border case.** The double-border case ($B_1 = 0$, $B_{k-1} = M$) is the most delicate, as the shift identity cannot be applied in either direction. We resolve it by reducing the equation $g(B) = -1$ to the vanishing of an explicit polynomial.

Notation 9.12. Write $\tilde{\sigma}(u) = \sum_{j=1}^{k-1} u^j$ for the partial geometric sum of u, and $\sigma(u) = 1 + \tilde{\sigma}(u) = \sum_{j=0}^{k-1} u^j$ for the full sum. These satisfy $\sigma(u) = (u^k - 1)/(u-1)$ when $u \neq 1$, and from $u^k = 2^k \cdot 3^{-k} = 2^S \cdot 2^{k-S} \cdot 3^{-k} \equiv 2^{k-S} \pmod d$ (since $2^S \equiv 3^k$), one obtains the identity

$$u^k \equiv 2^{k-S} = 2^{-M} \pmod d. \tag{15}$$

Lemma 9.13 (Iterated peeling)**.** *Suppose $B_1 = 0$ and $B_{k-1} = M$. Using (15), the boundary terms satisfy $u^{k-1} \cdot 2^M = u^{k-1} \cdot u^{-k} \cdot 2^k = u^{-1} \cdot 2^k \cdot 2^{-k} = u^{-1} \pmod d$, since $u^k = 2^{-M}$ gives $2^M = u^{-k}$ and thus*

$$u^{k-1-j} \cdot 2^M \equiv u^{-(j+1)} \pmod d \qquad \textit{for all } j \geq 0.$$

By fixing $B_1 = 0$ (contributing $u \cdot 2^0 = u$) and $B_{k-1} = M$ (contributing $u^{k-1} \cdot 2^M = u^{-1}$), then iteratively fixing the next innermost pair ($B_2 = 0, B_{k-2} = M$), etc., the equation $g(B) = -1$ reduces (for odd $k = 2n+3$) to a target value involving geometric sums P and Q:

$$P = \sum_{j=0}^{n} u^j = \frac{u^{n+1} - 1}{u - 1}, \qquad Q = \sum_{j=2}^{n+1} u^{-j}, \qquad \textit{target} = -(1 + P + Q).$$

The equation $g(B) = -1$ in the double-border case therefore reduces to $1 + P + Q \equiv 0 \pmod{d}$.

Definition 9.14 (Annihilation polynomial)**.** Define

$$F(u) \;=\; u^{2n+2} + u^{n+2} - 2u^{n+1} - u^n - 1, \tag{16}$$

where $n = (k-3)/2$ for odd k, so that $\deg(F) = k - 1$.

Proposition 9.15 (Polynomial criterion)**.** *For odd $k \geq 7$ with $\tilde{\sigma}(u) \neq 0$, the double-border equation $g(B) = -1$ has a solution if and only if*

$$F(u) \;\equiv\; 0 \pmod{d}. \tag{17}$$

Proof. The condition $1 + P + Q = 0$ is equivalent, after clearing denominators by multiplying by $(u-1) \cdot u^{n+1}$, to $F(u) = 0$. The identity $(1 + P + Q)(u - 1) = \psi(u^{n+1}) + \psi(u) - 2$, where $\psi(x) = x - x^{-1}$, can be verified algebraically. □

Theorem 9.16 (Integer evaluation)**.** *For odd $k = 2m + 1$ ($m \geq 3$), the integer*

$$F_{\mathbb{Z}} \;=\; 3^{k-1} \cdot F(2/3) \;=\; 4^m - 9^m - 17 \cdot 6^{m-1} \tag{18}$$

satisfies $F(u) \equiv 0 \pmod{d}$ if and only if $d \mid F_{\mathbb{Z}}$. Furthermore:

(a) *$F_{\mathbb{Z}}$ is always odd and coprime to 3;*
(b) *$F_{\mathbb{Z}} < 0$ for all $m \geq 2$;*
(c) *$F_{\mathbb{Z}} \equiv 3 \pmod{5}$ for all $m \geq 1$.*

Proof. Substituting $u = 2 \cdot 3^{-1}$ formally into $F(u)$ and multiplying by $3^{k-1} = 3^{2m}$ yields, after simplification, $4^m - 9^m - 17 \cdot 6^{m-1}$. Part (a): 4^m is even, 9^m is odd, $17 \cdot 6^{m-1}$ is even, so $F_{\mathbb{Z}}$ is odd; modulo 3, $4^m \equiv 1$, $9^m \equiv 0$, $17 \cdot 6^{m-1} \equiv 0$, giving $F_{\mathbb{Z}} \equiv 1 \pmod{3}$. Part (b): 9^m dominates 4^m for $m \geq 2$. Part (c): $4 \equiv 9 \pmod{5}$, so $4^m - 9^m \equiv 0$; $17 \cdot 6^{m-1} \equiv 2 \cdot 1 = 2$; hence $F_{\mathbb{Z}} \equiv -2 \equiv 3 \pmod{5}$. □

Remark 9.17 (Non-vanishing of $F_{\mathbb{Z}}$ modulo d). The condition $d \nmid F_{\mathbb{Z}}$ has been verified computationally for all odd $k \leq 10001$ (i.e., $m \leq 5000$). Beyond the computational range, a partial analytical argument applies. Since $d = 2^S - 3^k$ is odd and $F_{\mathbb{Z}}$ is odd (Theorem 9.16(a)), the equation $F_{\mathbb{Z}} = -n \cdot d$ for a positive integer n rearranges to $n \cdot 2^S = R(m)$ where $R(m)$ is a specific integer-valued expression. For $m \geq 4$, checking modulo 8 shows:

- $n = 1$: $R(m) \equiv 2 \pmod 8$, but $2^S \equiv 0 \pmod 8$ for $S \geq 3$. Contradiction.
- $n = 2$: $R(m) \equiv 7 \pmod 8$. Contradiction.
- $n = 3$: $R(m) \equiv 2 \pmod 8$. Contradiction.
- $n = 4$: $R(m) \equiv 5 \pmod 8$. Contradiction.

Hence $F_{\mathbb{Z}} \notin \{-d, -2d, -3d, -4d\}$ for $m \geq 4$. The first possible quotient is $n = 5$ (where $n \equiv 5 \pmod 8$ is required), but for typical values $|F_{\mathbb{Z}}|/d < 5$, so $d \nmid F_{\mathbb{Z}}$ follows. A complete analytical proof for all k remains open.

9.6. **Classification of $\tilde{\sigma} = 0$.** When $\tilde{\sigma}(u) = 0$, the constant sequence $B = (c, c, \ldots, c)$ gives $g(B) = \tilde{\sigma}(u) \cdot 2^c = 0$ for every c, so $0 \in \mathrm{Im}(g)$ but the target -1 is not directly produced. This case requires separate analysis.

Lemma 9.18. *$\tilde{\sigma}(u) = 0$ if and only if $d \mid (3^{k-1} - 2^{k-1})$.*

Proof. Since $\tilde{\sigma}(u) = (u^k - u)/(u - 1)$ when $u \neq 1$, and $u^k \equiv 2^{-M}$ by (15), the vanishing $\tilde{\sigma}(u) = 0$ is equivalent to $u^k = u$, i.e. $u^{k-1} = 1$. Substituting $u = 2 \cdot 3^{-1}$ gives $(2/3)^{k-1} \equiv 1 \pmod d$, i.e. $d \mid (2^{k-1} - 3^{k-1})$. □

Proposition 9.19 (Zsygmondy classification)**.** *Among all $k \geq 3$ for which $d = 2^S - 3^k$ is prime, $\tilde{\sigma}(u) = 0$ holds only for $k = 3$ and $k = 5$.*

Proof sketch. By Zsygmondy's theorem [23], the expression $3^{k-1} - 2^{k-1}$ has a primitive prime divisor for $k - 1 \geq 7$ (i.e. $k \geq 8$): a prime that divides $3^{k-1} - 2^{k-1}$ but does not divide $3^j - 2^j$ for any $j < k - 1$. For $d \mid (3^{k-1} - 2^{k-1})$ to hold with $d = 2^S - 3^k$, the divisor d would need to coincide with (or be composed of) such primitive prime divisors. A direct check for $k \leq 500$ confirms that only $k = 3$ ($d = 5$, $3^2 - 2^2 = 5$) and $k = 5$ ($d = 13$, $3^4 - 2^4 = 65 = 5 \cdot 13$) satisfy the condition. For $k \in \{6, 7\}$, the Zsygmondy exceptions are verified individually. □

Proposition 9.20 (Coupling $\tilde{\sigma} = 0$ and F)**.** *If $\tilde{\sigma}(u) = 0$ (i.e. $d \mid E_1$ where $E_1 = 3^{k-1} - 2^{k-1}$), then $F(u) \not\equiv 0 \pmod d$.*

Proof. The identity $F_{\mathbb{Z}} = -E_1 - 17 \cdot 6^{m-1}$ follows from $4^m = 2^{k-1}$ and $9^m = 3^{k-1}$ (with $m = (k-1)/2$). If $d \mid E_1$, then $F_{\mathbb{Z}} \equiv -17 \cdot 6^{m-1} \pmod d$. Since d is coprime to 6 and $d > 17$ for $k \geq 4$, we have $d \nmid 17 \cdot 6^{m-1}$, hence $F_{\mathbb{Z}} \not\equiv 0 \pmod d$. □

9.7. **Composite moduli and CRT reduction.**

Proposition 9.21 (CRT inequality)**.** *For any prime $p \mid d$,*

$$N_0(d) \;\leq\; N_0(p).$$

In particular, if $N_0(p) = 0$ for some prime factor p of d, then $N_0(d) = 0$.

Proof. If $\mathrm{corrSum}(A) \equiv 0 \pmod d$, then $\mathrm{corrSum}(A) \equiv 0 \pmod p$ for every $p \mid d$. Every solution modulo d is also a solution modulo p. □

Remark 9.22 (CRT anti-correlation). For composite $d = p_1 \cdots p_r$, it may happen that $N_0(p_i) > 0$ for every factor p_i individually, yet $N_0(d) = 0$. This occurs because the *same* composition A must satisfy $\mathrm{corrSum}(A) \equiv 0$ modulo *all* factors simultaneously: the structural relation $2^S \equiv 3^k \pmod{d}$ induces correlations between the reductions modulo distinct prime factors. This anti-correlation phenomenon has been verified by dynamic programming for all composite values of d with $k \leq 67$.

9.8. **Conditional main theorem.**

Theorem 9.23 (Blocking Mechanism, conditional)**.** *Assume the Generalized Riemann Hypothesis. Then for every $k \geq 3$,*

$$N_0(d) \;=\; 0,$$

where $d = 2^S - 3^k$ and $S = \lceil k \log_2 3 \rceil$. Combined with the Simons–de Weger bound (Theorem 4.1), this implies the nonexistence of nontrivial positive cycles in the Collatz dynamics.

Proof outline. We distinguish three cases for each k.

Case 1: d has a prime factor p with $N_0(p) = 0$. Then $N_0(d) = 0$ by Proposition 9.21. For $k \leq 67$, this is verified computationally by dynamic programming.

Case 2: d is prime and $\tilde{\sigma}(u) = 0$. This occurs only for $k = 3$ ($d = 5$) and $k = 5$ ($d = 13$) by Proposition 9.19. Both cases are resolved by direct computation: $N_0(5) = 0$ and $N_0(13) = 0$.

Case 3: d is prime and $\tilde{\sigma}(u) \neq 0$. By the four-case analysis (Definition 9.5):

- Interior and simple-border sequences: excluded by Corollary 9.8 and Lemmas 9.9–9.10, provided $\mathrm{ord}_d(2) > C$. Under GRH, Hooley's theorem [11] implies that 2 is a primitive root modulo d (or has index bounded by a small constant), so $\mathrm{ord}_d(2) \geq (d-1)/R$ for a bounded R. Since $C/d \to 0$ exponentially (by Stirling: $\log_2 C \approx 0.949 \cdot S$ while $d \approx 2^S$), $\mathrm{ord}_d(2) > C$ holds for all sufficiently large k, and the finitely many small k are verified computationally.
- Double-border sequences: excluded by Proposition 9.15 and Theorem 9.16, verified for $k \leq 10001$ ($F_{\mathbb{Z}} \not\equiv 0 \pmod{d}$). The coupling (Proposition 9.20) covers the $\tilde{\sigma} = 0$ cases.

Combining all cases, $N_0(d) = 0$ for every $k \geq 3$. □

Remark 9.24 (Role of GRH). The GRH is used *solely* to guarantee that $\mathrm{ord}_d(2)$ is large relative to C for primes d of the form $2^S - 3^k$. Without GRH, the argument is complete except for the bound on $\mathrm{ord}_d(2)$, which is a variant of Artin's primitive root conjecture restricted to a specific parametric family of primes. The remaining three gaps (finitude of $\tilde{\sigma} = 0$, non-vanishing of $F_{\mathbb{Z}}$, CRT anti-correlation for composite d) are supported by extensive computation but remain formally open.

9.9. **Computational verification.**

(V1) **Double-border non-vanishing**: $F_{\mathbb{Z}} \not\equiv 0 \pmod{d}$ for all odd $k \in [7, 10001]$ (4998 values) and all even $k \in [4, 1000]$ (499 values, no double-border solution found).

(V2) **Multiplicative order**: $2^C \not\equiv 1 \pmod{d}$ for all 19 primes d with $k \in [3, 10000]$. The quotient $(d-1)/\mathrm{ord}_d(2) \in \{1, 2, 3, 15\}$ for the 17 cases where $d-1$ could be factored.

(V3) **Coprimality**: $F_{\mathbb{Z}} \equiv 3 \pmod{5}$ for all m (Theorem 9.16(c)), and $p \nmid F_{\mathbb{Z}}$ for all m when $p \in \{7, 13, 19, 23, 29, 31, 43, 47, \ldots\}$ (proved by periodicity of $F_{\mathbb{Z}} \bmod p$). Only 8 *critical primes* $p \in \{11, 37, 53, 59, 109, 191, 283, 8363\}$ satisfy $p \mid F_{\mathbb{Z}}$ and $p \mid d$ simultaneously for some k, and at most one critical prime divides $\gcd(F_{\mathbb{Z}}, d)$ for each k (verified for $k \leq 10001$).

(V4) **CRT verification**: $N_0(d) = 0$ verified by dynamic programming for all $k \in [3, 67]$, covering all composite values of d in this range.

(V5) $\tilde{\sigma} = 0$ **classification**: $d \mid (3^{k-1} - 2^{k-1})$ with d prime occurs only for $k \in \{3, 5\}$ (exhaustive search for $k \leq 500$).

10. Conclusion and perspectives

10.1. **Assessment.** This work contains two complementary stages.

Stage I (unconditional, Sections 3–8) establishes that the evaluation map Ev_d is not surjective for $k \geq 18$ (Theorem 3.5), proving that at least one residue modulo d is unreachable. The gap between "some residue is omitted" and "the residue 0 is omitted" constitutes Hypothesis (H) (Section 7).

Stage II (conditional on GRH, Section 9) closes this gap via the Blocking Mechanism (Theorem 9.23): a four-case analysis of the Horner reformulation proves $N_0(d) = 0$ for all $k \geq 3$, assuming the Generalized Riemann Hypothesis. Without GRH, the argument depends on the multiplicative order $\mathrm{ord}_d(2)$ exceeding C for prime values of d, which is a variant of Artin's primitive root conjecture for the family $d = 2^S - 3^k$.

The character sum analysis (Sections 5–6) provides an independent analytical approach to the same gap by establishing:

(i) the minimal Parseval cost if a cycle exists (Theorem 5.4);
(ii) the exact decomposition of $T(t)$ into multiplicative characters (Theorem 6.2);
(iii) the peeling bound $N_0 \leq 0.63\, C$ (Lemma 7.3).

Quantitatively, the analysis yields a precise conditional criterion:

Proposition 10.1 (Theorem Q)**.** *Suppose that for every $k \geq 18$:*

(1) *for every prime $p \mid d(k)$,*

$$\left|\sum_{t=1}^{p-1} T(t)\right| \;\leq\; 0.041 \cdot C; \tag{19}$$

(2) *$d(k)$ has at least one prime factor $p > 1.041\,C(k)$.*

Then for every $k \geq 3$, no nontrivial positive cycle of length k exists.

Proof. For $k = 3, \ldots, 17$: $N_0(d) = 0$ by exhaustive verification (Section 8). For $k \geq 18$: condition (19) gives $|R(p)| \leq 0.041\,C/p$, hence $N_0(p) \leq 1.041\,C/p$ for each prime $p \mid d$. By the Chinese Remainder Theorem, $\mathrm{corrSum}(A) \equiv 0 \pmod{d}$ implies $\mathrm{corrSum}(A) \equiv 0 \pmod{p}$ for every prime $p \mid d$, so $N_0(d) \leq N_0(p)$ for each such p. In particular, $N_0(d) = 0$ whenever d has a prime factor $p > 1.041\,C$, which is precisely Condition (2). □

Remark 10.2 (Status of Condition (2)). Condition (2) requires that $d(k)$ possess a prime factor exceeding $1.041\,C(k)$. Since $C < d$ (Theorem 3.5) with $d/C \to \infty$ exponentially (Proposition 3.4), such a prime factor exists whenever $d(k)$ is itself prime or has a large prime component. However, numerical verification for $k \in [18, 100]$ shows that only 8 out of 83 values of k satisfy $\max_{p \mid d(k)} p > 1.041\,C(k)$. This condition is therefore *not* universally satisfied and constitutes a genuine hypothesis of Theorem Q. A refined argument exploiting the CRT inequality with *multiple* primes simultaneously—requiring $\prod_{p \mid d}(1 - N_0(p)/C) > 0$ rather than $N_0(p) = 0$ for a single p—could potentially relax Condition (2), but this is not pursued here.

Condition (19) is strictly weaker than Conjecture M: it requires only that the *sum* of exponential sums does not exceed 4.1% of C, rather than pointwise decay of each $|T(t)|$. Numerical evidence confirms it for all tested values ($k \leq 41$, all $p \mid d$). A systematic investigation (GPS methodology, 6 phases, ∼30 computational experiments) verifies Condition (19) for all $k \in [18, 28]$ and all primes $p \mid d(k)$ with $p \leq 50000$ (25 cases, 0 failures). The worst case is $k = 22$, $p = 7$, with ratio $|p \cdot N_0(p) - C|/C = 0.013$ (margin 3.2×). The distribution $N_r(p)$ is approximately constant on orbits of multiplication by 2 in $\mathbb{F}_p$ (max deviation 2.2%), and a decay rate $\sim k^{-6.3}$ is observed for $p = 7$ ($k = 22$ to 38).

Three-mesh net. A three-level proof structure covers all tested prime factors of $2^m - 1$ for $m = 1, \ldots, 100$ (168 primes, 0 failures):

(i) *Affine transport* (Section 5): covers all $p \leq 97$ unconditionally, via the commutator identity $[T_2, T_1] = \tau_{-1}$ and diameter $D(p) \leq 1.3 \log_2 p$;

(ii) *Convolution contraction*: for primes where $(p-1)\,\rho_p^{17} < 0.041$, the bound at $k = 18$ suffices (72 primes);

(iii) *Ghost fish*: for the remaining 72 primes (including Mersenne primes up to M_{89}), the convolution bound fails at $k = 18$ but no divisibility $p \mid d(k)$ occurs in the danger zone $[18, k_{\min}(p))$.

Direct verification for all known Mersenne primes M_q with $q \leq 127$ confirms the ghost property (up to 3738 values of k checked for M_{127}).

Two independent barriers protect larger Mersenne primes: a *size barrier* ($k \geq q \log_3 2 \approx 0.63\, q$, rigorous) and a *density barrier* (expected hits $E \leq C\, q^3/2^q \to 0$ super-exponentially, heuristic). For $q \geq 61$, the expected number of dangerous divisibilities is below 10^{-14}.

Junction geology (ord > 100**).** The three-mesh net was extended to primes with $\mathrm{ord}_p(2) > 100$ via a geological analysis of the Cunningham factorizations of $2^m - 1$ for $m = 101, \ldots, 250$ (all fully factored). A key structural quantity is $K_{\mathrm{MAX}} = \max_p k_{\min}(p)$, where $k_{\min}(p)$ is the smallest k such that $(p-1)\, \rho_p^{k-17} < 0.041$. Over all 168 primes with $\mathrm{ord}_p(2) \leq 100$:

$$K_{\mathrm{MAX}} = 63 \qquad (\text{attained by } M_{13} = 8191,\ \rho_{8191} = 0.763). \tag{20}$$

Since $63 \leq 68$, the junction between the Simons–de Weger bound ($k \leq 68$) and convolution ($k \geq K_{\mathrm{MAX}}$) has a six-rank overlap at $k \in [63, 68]$.

For primes with ord > 100, three complementary tests were performed:

(a) *Category A* ($p < 20000$, 89 primes): exact computation gives $\rho_{\max} = 0.280$, all pass;
(b) *Danger primes* (11 primes with $p > 10^8$ and $\rho > 0.5$): ghost fish analysis shows that *none* divides any $d(k)$ for $k \in [18, 300]$;
(c) *Gap scan* ($k \in [69, 120]$, 242 primes from factoring $d(k)$): all with computable ρ satisfy $\rho < 0.23$, passing by margins exceeding 10^{40}.

The phenomenon underlying (b) is a *fish-tunnel incompatibility*: primes with large ρ (due to $p \gg m^2$) have structural constraints (large ghost fish period P) that prevent them from dividing $d(k)$, while primes that *do* divide $d(k)$ have large ord and correspondingly small ρ.

Combining these results, the estimated feasibility of Condition (19) for all $k \geq 18$ rises to $4.75/5$.

Regime analysis via exponential sum bounds. A deeper analysis decomposes the verification of Condition (19) according to the exponential sum parameter

$$\rho(p) = \max_{c \neq 0} \frac{\left|\sum_{h \in \langle 2 \rangle} \omega^{ch}\right|}{m}, \qquad m = \mathrm{ord}_p(2), \quad \omega = e^{2\pi i/p}.$$

Regime A ($p < m^4$). The bound of Di Benedetto, Garaev, Garcia, González-Sánchez, Shparlinski, and Trujillo [6] gives $\rho \leq C_1 \cdot m^{-31/2880}$, yielding a uniform critical index $k_{\mathrm{crit}}(p) \leq K_A \approx 400$. Computational verification of Condition (19) for all $k \in [69, 500]$ and all prime factors of $d(k)$ in Regime A was performed (116/132 cases fully verified, 0 failures, 16 timeouts from large cofactors).

Regime B ($p \geq m^4$). Only the trivial bound $\rho \leq 1 - 1/m$ is available. Let $n_3 = \min\{j \geq 1 : 3^j \in \langle 2 \rangle \bmod p\}$. Then $p \mid d(k)$ forces $n_3 \mid k$ (first filter) and a Beatty-type congruence $\lceil n_3 j \theta \rceil \equiv Lj \pmod{m}$ (second

filter), where $\theta = \log_2 3$ and $L = \log_2(3^{n_3}) \bmod m$. In the generic case $n_3 = (p-1)/m$ (which covers 51% of observed divisibility instances for $k \in [69, 150]$), one shows $k_{\text{crit}}(p) < n_3$ for $m \geq 4$, so that *no* multiple of n_3 falls in the danger zone; hence no prime in this case divides any $d(k)$ in the critical range. In the general case ($3 \notin \langle 2 \rangle$, $n_3 < (p-1)/m$), the number of multiples of n_3 in $[69, k_{\text{crit}}]$ satisfying the Beatty congruence is bounded by $N(p, k_{\text{crit}}) \leq \lfloor 3\ln(p)/n_3 \rfloor + 1$, which is $O(\ln p/n_3)$. When $n_3 > 3m \ln p$, one obtains $N = 0$ directly (since $k_{\text{crit}} < n_3$). For small n_3 (e.g. $n_3 = 2$), the bound can reach $O(\ln m)$, so the Beatty filter alone does not close this case. Empirically, however, Regime B is *empty* among all 474 observed divisibility pairs (k, p) for $k \in [18, 150]$: every prime factor of $d(k)$ satisfies $p < m^4$ (284 pairs for $k \in [69, 150]$, plus 190 additional pairs for $k \in [18, 67]$ via complete factorization of $d(k)$ by ECM; see Section 8).

Combining the regime analysis with the junction geology and the complete factorization of $d(k)$ for $k \in [18, 67]$ (190 additional prime factors, all Regime A; see Section 8), the estimated feasibility of Condition (19) for all $k \geq 18$ rises to 4.90/5. The residual gap is the case $n_3 \leq 3m \ln p$ in Regime B, where $N = O(\ln p/n_3)$; this case is empirically absent over 474 tested pairs.

Quantitative effectivisation (L12). A systematic enumeration of all Regime B primes (factors p of $2^m - 1$ with $p \geq m^4$, for $m \leq 130$) identifies exactly 20 such primes: 7 Mersenne primes ($M_{17}, M_{19}, M_{31}, M_{61}, M_{89}, M_{107}, M_{127}$) and 13 composite Mersenne factors. For each, we verify Condition (19) via two complementary mechanisms:

(i) *Spectral bound*: for 6 primes with $m \geq 47$, exact FFT computation of $\rho(p)$ yields $k_{\text{crit}}(p) < 69$, so Condition (19) holds automatically for all $k \geq 69$. Representative values: $\rho(M_{17}) = 0.8135$ ($k_{\text{crit}} = 89.6$, resolved by (ii)); $\rho(13264529) = 0.5409$ ($k_{\text{crit}} = 48.9 < 69$).

(ii) *Non-divisibility*: for all 20 primes, $p \nmid d(k)$ for every $k \leq 10^5$. This is verified by modular arithmetic: $2^{\lceil k\theta \rceil} \not\equiv 3^k \pmod{p}$ for all relevant k.

Result: 20/20 Regime B primes satisfy Condition (19). The cascade of filters (modular filter $n_3 \mid k$, Beatty congruence, spectral bound) reduces the expected number of exceptions to $N_2 < 1$ for 15/20 primes, with the remaining 5 resolved by exact ρ-computation or non-divisibility.

The Konyagin [12] bound $\rho(p) \leq \exp(-c \cdot (\log p)^{1/3})$ for $|H| \geq p^{1/4+\varepsilon}$ (with $c > 0$ non-explicit) would close the entire Regime B gap if $c \geq c_{\min}$, where

(21)
$$c_{\min} = \max_{k \geq 69} \frac{\ln d(k) + 3.19}{(k-17) \cdot (\ln d(k))^{1/3}} = 0.3572 \qquad \text{(attained at } k = 69\text{)}.$$

The function $c_{\text{needed}}(k)$ is strictly decreasing for $k \geq 69$ (verified numerically). Thus, an effective version of the Bourgain–Glibichuk–Konyagin [9] theorem with $c \geq 0.36$ would yield an unconditional proof of Condition (19) for all Regime B primes.

Extended effectivisation (L13). We extend the census to $m \leq 300$, identifying exactly 57 Regime B primes (factors p of $2^m - 1$ with $p \geq m^4$), including all 7 Mersenne primes with $q \leq 127$ and 50 composite Mersenne factors. For each of these 57 primes, we verify:

(i) *Non-divisibility*: for 56/57 primes, $p \nmid d(k)$ for all $k \in [69, 10^6]$;
(ii) *Safe divisibility*: the unique exception (M_{17}, first hit at $k = 7710$) satisfies $k_0 = 7710 \gg k_{\text{crit}}(M_{17}) = 264$, so $H(7710)$ holds by the Regime A factors of $d(7710)$.

Furthermore, for $k \in [69, 200]$, we verify that *every* prime factor of $d(k)$ is Regime A with margin $\geq 91\%$ (maximum ratio $p/m^4 = 0.086$, attained at $p = 7$, $m = 3$). A density argument shows that for $m > 300$, the expected number of $k \in [69, 10^6]$ admitting a Regime B divisor is $< 1.2 \times 10^{-4}$.

Proposition 10.3 (Effective Regime B vacuity)**.** *Let $\mathcal{B}(M) = \{p$ prime : $p \mid (2^m - 1),\ p \geq m^4,\ 17 \leq m \leq M\}$. Then:*

(a) $|\mathcal{B}(300)| = 57$ and no Regime B prime exists for $m \leq 16$ (since $2^m - 1 < m^4$ for $2 \leq m \leq 16$);
(b) for all $p \in \mathcal{B}(300)$ and all $k \in [69, 10^6]$, either $p \nmid d(k)$, or the first k_0 with $p \mid d(k_0)$ satisfies $k_0 > k_{\text{crit}}(p)$;
(c) for all $k \in [69, 200]$, every prime $p \mid d(k)$ satisfies $p < m^4$ (Regime A), with $\max_{p,k}(p/m^4) < 0.087$.

Proof. Parts (a)–(c) are established by direct computation (`phase_c_structural_proof.py`, `phase_d_formal_proof.py`). Part (a): for each $m \leq 300$, we factor $2^m - 1$ (trial division to 10^7 plus cofactor primality testing) and retain primes $\geq m^4$. Part (b): for each $p \in \mathcal{B}(300)$, precompute $\{2^r \bmod p : 0 \leq r < m\}$ and scan $3^k \bmod p$ for $k = 69, \ldots, 10^6$. Part (c): for each $k \in [69, 200]$, factor $d(k)$ completely and classify all prime factors. □

Combined with Theorem 4.2, this yields Condition (19) unconditionally for $k \in [69, 10^6]$, and for all $k \geq 69$ under the *Regime A Universality* hypothesis ($p \mid d(k) \Rightarrow p < (\text{ord}_2(p))^4$).

CRT bypass of Regime B (L14). The Chinese Remainder Theorem implies a decisive simplification: since $N_0(d) \leq N_0(p)$ for every prime $p \mid d$, verifying Hypothesis (H) requires $N_0(p) = 0$ for only *one* prime factor of $d(k)$, not all of them. Combining this with the mechanism of Theorem Q (Proposition 10.1): if Condition (19) holds for a single prime $p \mid d(k)$ with $p > 1.041\, C(k)$ (i.e., Conditions (1) and (2) of Theorem Q are met for that prime), then $N_0(p) \leq 1.041\, C/p < 1$, hence $N_0(p) = 0$ and $N_0(d) = 0$.

Proposition 10.4 (One Good Prime Suffices)**.** *For every $k \geq 18$:*

(a) $d(k)/C(k) \geq 1.58$ (verified for $k \leq 500$; $d/C \to \infty$ exponentially by Proposition 3.4);
(b) no Regime B prime divides $d(k)$ for $k \in [69, 10^4]$ (Proposition 10.3);
(c) every prime below 10^5 is Regime A (since the smallest Regime B prime is $M_{17} = 131071$);
(d) for each tested k, every prime factor of $d(k)$ found by factorisation is Regime A.

In particular, to prove $H(k)$ for all $k \geq 69$, it suffices to verify Condition (19) for a single *Regime A prime $p \mid d(k)$ with $p > 1.041\, C(k)$. Regime B primes are entirely irrelevant.*

Proof. Part (a): direct computation for $k \leq 500$; asymptotic growth by Proposition 3.4. Part (b): Proposition 10.3, noting the unique hit $M_{17} \mid d(7710)$ is dominated by larger Regime A factors of $d(7710)$. Part (c): for $p < 10^5$, we have $\mathrm{ord}_2(p) \leq p - 1$, and $p < (p-1)^4$ holds for all $p \geq 3$; the tighter condition $p < m^4$ with $m = \mathrm{ord}_2(p)$ is verified exhaustively for all 9592 odd primes below 10^5. Part (d): deep factorisation of $d(k)$ for 13 values with no small factor $< 10^3$ confirms all prime factors are Regime A (`phase_e_one_good_prime.py`).

The CRT inequality $N_0(d) \leq N_0(p)$ then gives the conclusion: Condition (19) is needed only for the single Regime A prime selected above. □

10.1.1. *Structural lemmas for Mersenne primes.* For Mersenne primes $M_q = 2^q - 1$, the subgroup $\langle 2 \rangle = \{2^0, 2^1, \ldots, 2^{q-1}\}$ has a particularly rigid binary structure that yields exact combinatorial results.

Lemma 10.5 (Additive energy of $\langle 2 \rangle$ for Mersenne primes)**.** *For $p = M_q$ with $q \geq 3$:*

(i) $|\langle 2 \rangle + \langle 2 \rangle| = q(q+1)/2$;
(ii) $E(\langle 2 \rangle) = 2q^2 - q$.

Proof. Since $2^j < 2^q - 1 = p$ for $0 \leq j \leq q-1$, elements of $\langle 2 \rangle$ are unreduced powers of 2. The sumset decomposes into "doubles" 2^{i+1} (including $2^q \equiv 1$) giving q elements in $\langle 2 \rangle$, and "distinct pairs" $2^i + 2^j$ ($i < j$), each with exactly two bits set in binary, hence pairwise distinct and distinct from elements of $\langle 2 \rangle$. Since $2^i + 2^j < 2^{q-1} + 2^{q-2} < p$, no reduction occurs. Total: $q + \binom{q}{2} = q(q+1)/2$.

For the energy: doubles contribute $q \times 1^2 = q$ (each sum 2^{i+1} has one representation (i,i)), while distinct pairs contribute $\binom{q}{2} \times 2^2 = 2q(q-1)$ (each $2^i + 2^j$ has representations (i,j) and (j,i)). Total: $E = q + 2q(q-1) = 2q^2 - q$. □

Lemma 10.6 (Structural n_3 bound for Mersenne primes)**.** *For $p = M_q$ with $q \geq 5$: $n_3(M_q) \geq \lceil q \cdot \log_3(2) \rceil = \lceil q/\theta \rceil$, where $\theta = \log_2(3)$.*

Proof. For $1 \le j \le \lfloor q/\theta \rfloor$, we have $3^j < 2^q = p+1$, so $3^j \bmod p = 3^j$ (unreduced). Since 3^j is odd and > 1 for $j \ge 1$, while every element of $\langle 2 \rangle = \{1, 2, 4, \ldots, 2^{q-1}\}$ with exponent ≥ 1 is even, the only possible match is $3^j = 1 = 2^0$, which fails for $j \ge 1$. Hence $3^j \notin \langle 2 \rangle$ for all $j \in [1, \lfloor q/\theta \rfloor]$, giving $n_3 > \lfloor q/\theta \rfloor$. Since q/θ is irrational (θ is transcendental), $n_3 \ge \lfloor q/\theta \rfloor + 1 = \lceil q/\theta \rceil$. □

Corollary 10.7. *For $q \ge 110$, we have $n_3(M_q) \ge 70 > 69$, so the modular filter $n_3 \mid k$ excludes $k = 69$ entirely from the divisibility candidates.*

Spectral concentration: a negative result. The *spectral concentration ratio* $C(q) := \max_{t \ne 0} |S_t|^4 / \mathrm{avg}_{t \ne 0} |S_t|^4$ satisfies $C(q) \sim q^2/2$ as $q \to \infty$ (verified numerically for $q \le 19$). Combined with Lemma 10.5, $\mathrm{avg}|S_t|^4 \approx 2q^2$, this gives $\rho(M_q)^4 \sim C(q) \cdot 2q^2/q^4 \to 1/2$, i.e., $\rho(M_q) \to 2^{-1/4} \approx 0.8409$. The convergence is super-polynomial, with $|1 - 2\rho^4|$ decreasing faster than any power of $1/q$.

This establishes that **no moment-based bound** (4th, 6th, or any $2k$-th moment) can prove $\rho \to 0$ for Mersenne primes. The spectral concentration is inherent to the binary structure of $\langle 2 \rangle$.

Divisibility verification. Direct computation confirms that $M_{17} \mid d(7710)$ (with $n_3 = 7710$, $\beta_0 = 15$, and the Beatty congruence $\lceil 7710\theta \rceil \equiv 15 \pmod{17}$ satisfied). However, $k = 7710 \gg k_{\mathrm{crit}} \approx 97$, so $(p-1) \cdot \rho^{k-17} \approx 10^{-684} \ll 0.041$. Similarly, $M_5 \mid d(48)$, $M_7 \mid d(90)$, all safely outside the danger window. *No divisibility occurs within the danger window* $[18, k_{\mathrm{crit}}]$ for any Mersenne prime tested.

10.2. **Open difficulties.** Proposition 7.9 shows that no purely spectral method (moments of T) can close the gap in the regime $p \sim C^{1+o(1)}$. The structural analysis of Section 10.1.1 makes this precise: $\rho(M_q) \to 2^{-1/4}$ (conjectured), so the spectral gap saturates at a *positive constant*. Closing the formal gap requires tools beyond moment methods.

Identified approaches are:

(a) *Effectivisation of Konyagin's bound*: extracting $c \ge 0.36$ from the Bourgain–Glibichuk–Konyagin [9] proof;
(b) *Diophantine arguments*: proving $n_3(M_q) \ge 4q$ for large Mersenne primes (current structural bound gives only $n_3 \ge 0.631q$, a factor $6.3\times$ gap);
(c) the *additive energy* of the subset $\{2^0, \ldots, 2^{S-1}\}$, whose exact value $E(\langle 2 \rangle) = 2q^2 - q$ for Mersenne primes (Lemma 10.5) provides structural constraints (cf. Applegate and Lagarias [1]);
(d) *Baker's theory of linear forms in logarithms*, which controls $|\lceil k\theta \rceil \cdot \log 2 - k \log 3|$ but not the residue $\lceil k\theta \rceil \bmod q$ required for divisibility.

A complementary difficulty, identified by the three-mesh net analysis, is the *ghost fish gap*: for an undiscovered Mersenne prime M_q

with $q > 127$, proving that $3^k \notin \{2^0, \ldots, 2^{q-1}\} \pmod{M_q}$ for all k in the danger zone $[k_{\text{size}}, k_{\min})$. The expected number of exceptions is super-exponentially small ($\leq C\, q^3/2^q$), but a rigorous proof would require either effective equidistribution of $3^k \bmod M_q$, or extension of the Cunningham factorization tables beyond $m = 100$.

The junction geology (Section 10) narrows this gap considerably. The key quantity $K_{\text{MAX}} = 63$ shows that for $k \geq 63$, *all* primes with $\text{ord}_p(2) \leq 100$ satisfy Condition (19). The residual gap reduces to: primes with ord > 100 that divide $d(k)$ for some $k > 120$ and whose ρ exceeds a computable threshold. Empirically, no such prime has been found among 242 tested factors of $d(k)$ for $k \in [69, 120]$, and the 11 known primes with $\rho > 0.5$ and ord > 100 are ghost fish (absent from all $d(k)$).

The regime analysis further narrows the gap: Regime A is closed by [6] combined with finite verification, while Regime B is addressed by a Beatty counting argument and quantitative effectivisation. The complete factorization of $d(k)$ for all $k \in [18, 67]$ (Section 8) shows that *every* prime factor in this range is in Regime A (190/190, 0 in Regime B), confirming that the Regime B case does not arise in the junction overlap zone. In the generic case ($n_3 = (p-1)/m$), one obtains $N = 0$. The formal lacuna is the case $n_3 \leq 3m \ln p$ in Regime B, where the equidistribution bound gives $N \leq \lfloor 3\ln(p)/n_3 \rfloor + 1 = O(\ln p/n_3)$. All 20 concrete Regime B primes (for $m \leq 130$) satisfy Condition (19), and the required Konyagin constant is modest ($c \geq 0.36$, see (21)). Moreover, the complete factorization of $d(k)$ for $k \in [18, 67]$ reveals that Regime B is absent in this range as well (0/190), extending the empirical observation from 0/284 ($k \in [69, 150]$) to 0/474 ($k \in [18, 150]$). Closing this gap unconditionally would require an effective version of the Bourgain–Glibichuk–Konyagin [9] sum-product estimate for $\langle 2 \rangle \bmod p$ with explicit constant $c \geq 0.36$.

These approaches connect to three precise open conjectures identified in the course of this work:

- *Horner equidistribution* (Conjecture 22.3): there exists $\delta > 0$ such that $|N_r(p) - C/p| \leq C \cdot p^{-1/2-\delta}$ for every $r \in \mathbb{F}_p$;
- *Strong spectral gap* (Conjecture Δ'): the effective spectral gap of the Horner walk satisfies $\Delta_{\text{eff}} \geq \delta_1 \cdot S/k$ for some $\delta_1 > 0$;
- *Uniform proportion* (Conjecture PU): the ordering constraint on compositions is asymptotically independent of the residue class modulo p.

Under Conjectures Δ' and PU jointly, one obtains a conditional chain: low additive energy of $H \Rightarrow$ spectral mixing $\Rightarrow$ equidistribution $\Rightarrow$ CRT exclusion $\Rightarrow$ no cycles. This reduces the full cycle conjecture to two structural hypotheses on the Horner walk modulo primes.

10.3. **Scientific transparency.** We believe in transparent science. A proof attempt via Baker's theory (Kolmogorov–Baker bounds) was **rejected** after self-audit. Details are available in the accompanying online repository.

Acknowledgements

The author thanks the anonymous contributors to the online discussions that helped refine this work, and acknowledges the Lean 4 community for the proof assistant infrastructure. This research was conducted independently without institutional funding.

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